Numerical Methods: x = g(x) Method

Questions 1–18

Question 1. Show that the equation $x^2 + x - 4 = 0$ may be arranged in to each of the following:

a) $x = \sqrt{4 - x}$

b) $x = \frac{4}{x+1}$

Question 2. Show that the equation $x^3 + 2x - 5 = 0$ may be arranged in to each of the following:

a) $x = \sqrt[3]{5 - 2x}$

b) $x = \frac{5}{x^2+2}$

Question 3. Given that $f(x) = x^3 - x^2 - 2x - 3$

a) Show that the equation $f(x) = 0$ may be written in the form $x = \sqrt{\frac{3+2x}{x-1}}$

b) Use your answer to part a) to write a recurrence relation for finding a solution to $f(x) = 0$ then starting with $x_1 = 2$, find a solution to 4 decimal places.

Question 4. Given that $f(x) = x^2 \ln x - 5x$

a) Show that $f(x) = 0$ has a solution between $x = 3$ and $x = 4$

b) Show that the equation $f(x) = 0$ may be written in the form $x = \sqrt{\frac{5x}{\ln x}}$

c) Use your answer to part b) to write a recurrence relation for finding a solution to $f(x) = 0$ then starting with $x_1 = 3$, find a solution to 4 decimal places.

Question 5. Prerequisite knowledge for Q5 b): Radians For each of the following recurrence relations, find x_2, x_3 and x_4 giving your answer to 4 decimal places:

a) $x_{n+1} = \sqrt{x_n + 3}$ where $x_1 = 5$

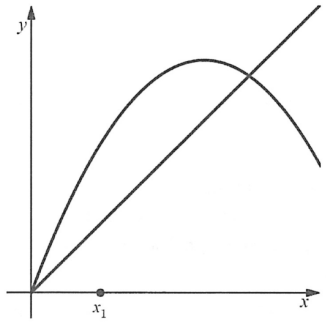
b) $x_{n+1} = 2\sin(x_n)$ where $x_1 = 1$ (x_n is measured in radians)

c) $x_{n+1} = 8 - \frac{1}{10}e^{0.4x_n}$ where $x_1 = 0$

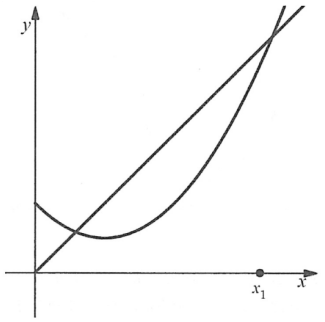
d) $x_{n+1} = 0.1x_n^3 - 0.5x_n + 2$ where $x_1 = 2$

Question 6. For each graph below, draw a diagram to show the convergence or divergence cobweb or staircase starting at the indicated point, labelling each of x_2, x_3 and x_4.
Identify whether it is a convergent or a divergent cobweb or staircase diagram.

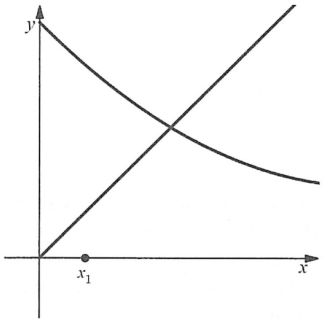
a)



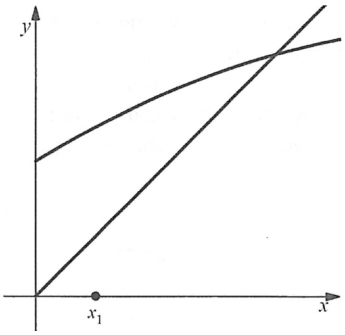
b)



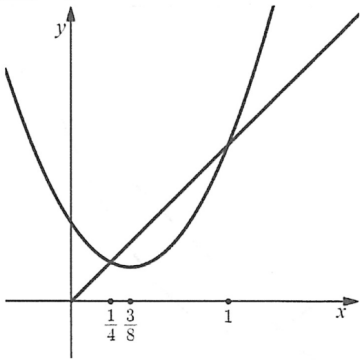
c)



d)



Question 7. The diagram shows $y = 2x^2 - 1.5x + 0.5$ and $y = x$ on the same graph. They intersect when $x = \frac{1}{4}$ and $x = 1$.



Describe what happens in terms of a cobweb or staircase diagram for each of the following starting points:

a) $-\frac{1}{4} < x_1 < \frac{1}{4}$
b) $x_1 = \frac{1}{4}$
c) $\frac{1}{4} < x_1 < 1$
d) $x_1 = 1$
e) $x_1 > 1$

Question 8. Prerequisite knowledge for Q8 b): Radians Each of the following iterative formulae has a sequence which converges to a limit $x = \alpha$. In each case, deduce an interval of width 0.001 in which α lies:

a) $x_{n+1} = 2 + \ln x_n$ where $x_1 = 3$
b) $x_{n+1} = \cos^2(x_n) + 1$ where $x_1 = 1.15$ (x_n is measured in radians)
c) $x_{n+1} = \frac{3}{2}x_n - \frac{1}{10}x_n^4$ where $x_1 = 1$

Question 9. An iterative formula for finding root α is given by $x_{n+1} = \mathrm{g}(x_n)$. By considering $\mathrm{g}'(\alpha)$ in each of the following cases, explain whether the iterative formula will converge or diverge to the given root (given to 3 decimal places) from a point considered close to the root.

a) $x_{n+1} = x_n^5 - 2x_n^4 - 5$ where $\alpha = 2.273$
b) $x_{n+1} = e^{x_n} - 2$ where $\alpha = 1.146$
c) $x_{n+1} = e^{x_n} - 2$ where $\alpha = -1.841$

Question 10. Harry is trying to find the values of the three roots of the equation $y = x^3 - 3x^2 - x + 4$. He rearranges the equation into four of the possible iterative formula $x_{n+1} = \mathrm{g}(x_n)$ to find a solution α, including finding an initial point:

$$\begin{aligned}
&\text{A}: \quad x_{n+1} = x_n^3 - 3x_n^2 + 4 && x_1 = 2 \\
&\text{B}: \quad x_{n+1} = \sqrt[3]{3x_n^2 + x_n - 4} && x_1 = 2.8 \\
&\text{C}: \quad x_{n+1} = \frac{x_n - 4}{x_n^2 - 3x_n} && x_1 = 1.2 \\
&\text{D}: \quad x_{n+1} = -\sqrt{\frac{x_n^3 - x_n + 4}{3}} && x_1 = -1
\end{aligned}$$

a) Find the three roots using the iterative formulae B, C and D at the given initial points above, giving your answers to 2 decimal places.
b) By considering $\mathrm{g}'(\alpha)$ on iterative formula A, explain why this iterative process does not converge to a root for each of the roots that you found in part a).

Question 11.

a) Sketch the graphs $y = \frac{1}{x}$ and $y = \sin x$ to show that the equation $\frac{1}{x} = \sin x$ has exactly one solution for $0 < x < \frac{\pi}{2}$

b) Show that the solution to the equation lies between $x = 1$ and $x = 1.2$

c) By using a suitable iterative formula, obtain an approximation solution to $\frac{1}{x} = \sin x$ to 4 decimal places.

Question 12.

a) Sketch the graphs $y = x$, $y = e^{x-2}$ and $y = 2 + \ln x$ on the same axes, for $0 < x < 4$

b) Explain why the solutions to the equation $x = e^{x-2}$ are also solutions to the equation $e^{x-2} = 2 + \ln x$

Question 13. The number of tickets sold N for an event in the first 48 hours of sales is modelled by $N = 3t - e^{0.1t}$ where t is the time in hours passed. When the number of tickets sold drops to zero and $t = T$

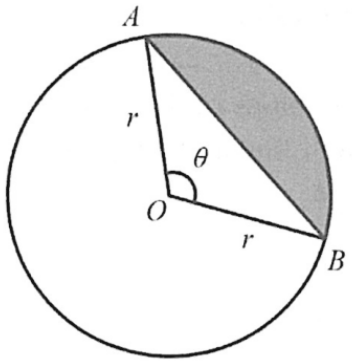
a) Show that T satisfies $T = 10\ln(3T)$

b) Write a recurrence relation to find the solutions to $T = 10\ln(3T)$ and starting with $T_1 = 48$, find a solution to the equation to 3 decimal places.

c) Hence state the number of hours that have passed, to the nearest hour, when the number of tickets being sold is zero.

Question 14. Prerequisite knowledge for Q14: Radians

A circle with centre O has a chord such that the angle $AOB = \theta$ radians where $0 < \theta < \pi$



The area of the segment cut off by AB is $\frac{1}{6}$ of the area of the circle.

a) Show that $\theta - \sin\theta = \frac{\pi}{3}$

The solution lies between $\theta = 1$ and $\theta = 3$

b) Using an iterative method, find the solution correct to 2 decimal places.

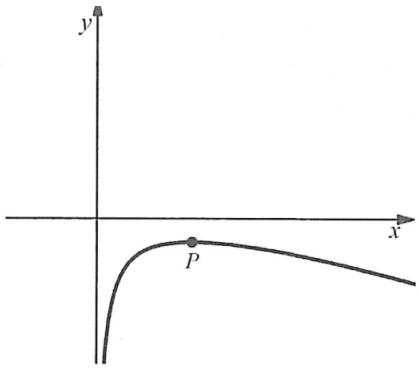
Question 15. Giving your answers to 3 decimal places, use an iterative method to find the x-coordinates of the points of intersection of:

a) $y = x$ and $y = \cos x$
b) $y = x^2$ and $y = x \log_3(0.5x + 1)$

Question 16. Prerequisite knowledge for Q16: Inverse Trigonometric Functions

a) Sketch the graphs of $y = 3\arctan(2x)$ and $y = x^2$ on the same axes for $-4 \leq x \leq 4$
b) Show that there is a solution to $x^2 - 3\arctan(2x) = 0$ between $x = 1.8$ and $x = 2.2$
c) By using a suitable iterative formula, obtain an approximation solution to $x^2 - 3\arctan(2x) = 0$ to 3 decimal places.

Question 17. Prerequisite knowledge for Q17: Differentiation of $\ln x$ The diagram shows the graph of $y = -\frac{2x^2+24}{3\sqrt{x}} - 4\ln x$. The stationary point of the curve is labelled P.



a) Show that the point P satisfies $0 = \sqrt{x} - \frac{1}{4}x^2\sqrt{x} - x$
b) Use the iteration formula $x_{n+1} = \sqrt{x_n} - \frac{1}{4}x_n^2\sqrt{x_n}$ starting at $x_1 = 1$ to find the x-coordinate of point P to 2 decimal places.

Question 18. Prerequisite knowledge for Q18: Double Angle Formulae An approximate solution to an equation can be found using the iterative formula $x_{n+1} = \frac{1}{4}\arccos x_n$.

a) By using $x_1 = 0.3$, find x_2, x_3 and x_4 giving your answers to 4 decimal places.
b) Show that the iterative formula may be used to find a solution to $x - 1 + 8\sin^2 x \cos^2 x = 0$